
Preface of UniReps: the Third Workshop on Unifying Representations in Neural Models

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Abstract

Discover why, when and how distinct learning processes yield similar representations, and the degree to which these can be unified.

<https://unireps.org>

Introduction

We are pleased to introduce this preface to the proceedings of the 3rd edition of the UniReps workshop on unified representations, held at NeurIPS 2025 in San Diego, USA. You can find the full [recordings of the workshop here](#). For the proceedings of the previous editions, please refer to [10] and [7]. In the following sections, we summarize the workshop’s organization, present key statistics, and discuss its future direction.

Workshop Summary

Neural models tend to learn similar representations when subject to similar stimuli. This behavior has been observed both in biological [11, 19] and artificial settings [22, 18, 23, 13]. The word *similar* here plays a fundamental role: under different conditions and assumptions on the observed data and the neural model (e.g., two distinct individuals exposed to the same stimulus [27] or different initializations of the same neural architecture [36]), inner representations of distinct models can be

related to each other, for instance up to a linear transformation [29]. Similar representations can emerge in distinct models when data are acquired from multiple modalities, for instance textual and image representations of the same entity [26], or in a multiview setting [34] where data in a single modality are acquired under different conditions.

The emergence of similar representations is a ubiquitous but elusive phenomenon, which is igniting a growing interest in the fields of Neuroscience, Artificial Intelligence and Cognitive Science. It is central to the question of understanding how mechanisms relate between different neural networks (NNs), and between NNs and brains. The goal of this workshop was to facilitate a rich exchange of ideas and insights among experts in these fields, by addressing the following points:

- (When): To explore the specific conditions under which these similarities emerge in different neural models. Modelling the transformations, symmetries and invariances between similar representations is key to measure if these can be unified [18, 17, 20].
- (Why): To investigate the underlying causes of similarities between neural representations, focusing on both artificial and biological models, as well as across them. Promising directions include analyzing the learning dynamics of neural models [1, 3, 31], studying model identifiability in the functional and parameter space [32, 14, 29, 15] and investigating the relations between different local minima reached by the optimization process [9, 8, 21].
- (What for): To explore and showcase applications in modular deep learning ranging from model merging [2], reuse [6, 16] and stitching [4, 25] to efficient strategies for fine-tuning and knowledge transfer between models [37] even in out-of-distribution settings [28], or to exploit cross-domain representation similarities (e.g. comparisons between images and neural data [33]).

The workshop aimed to provide an exciting, timely, and diverse environment for discussing theoretical findings, empirical evidence, and practical applications of the emergence of similar representations across models, benefiting from the cross-pollination of different fields (ML, Neuroscience, Cognitive Science) to foster the exchange of ideas and encourage collaborations. To this end, we invited contributions on the following topics:

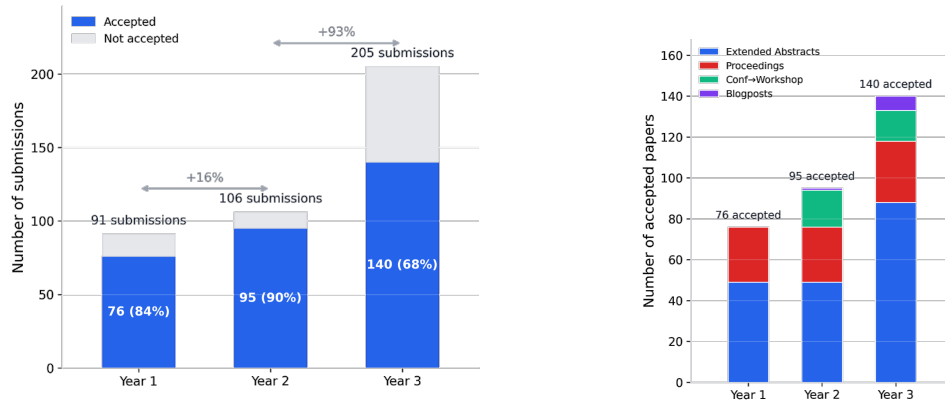
- Model merging, stitching and reuse [2]
- Symmetry and equivariance in NNs [12]
- Identifiability in neural models [29]
- Synergy of biological & artificial NNs [5]
- Learning dynamics [31]
- Multiview representation learning [34]
- Convergent learning [30, 13]
- Linear mode connectivity [9]
- Similarity based learning [35, 38]
- Multimodal learning [26]
- Representational alignment [24]

Workshop Format We designed a dynamic workshop program integrating (i) *invited talks* (ii) with a *mentorship panel discussion*, (iii) a *poster session* and (iv) a *social event*. In the *mentorship panel discussion*, we gathered renowned experts from the fields of AI, Neuroscience, and Cognitive Sciences to engage in a dynamic round table discussion on key topics explored in the workshop. We aimed to establish a cohesive understanding of the emergence of similar representations in neural models and pave the way for a new interdisciplinary community and research area, fixing the relevant research questions to be addressed. For the *mentoring panel discussion*, we invited Ahmad Beirami (Google DeepMind), Sara Hooker (Cohere Labs) and Meenakshi Koshla (MIT) to discuss topics related to peer reviewing, with a focus on how to navigate the current challenges in peer reviewing and publishing in interdisciplinary fields. The *poster session* provided all authors of accepted papers the chance to showcase their work, share findings, and conduct meaningful discussions among peers. Finally, two *social events* took place after the workshop, one immediately after the workshop and the other the next day in collaboration with the NeurReps workshop, fostering informal connections among participants and favoring the establishment of long term relationships and collaborations within the workshop community.

Multiple submission tracks Submissions to the workshop were organized in two technical tracks, both requiring novel and unpublished results: we received 88 submissions for the *extended abstract* track, which addresses early-stage results, insightful negative findings, position papers, and 30 submissions for the *proceedings* track, which address complete papers to be published in a dedicated

Schedule			
08:15 AM	Opening Remarks	12:15 PM	Lunch
08:30 AM	Invited Talk: R. Yu	1:30 PM	Invited Talk: R. Balestriero
09:00 AM	Invited Talk: N. Kriegeskorte	2:00 PM	Contributed Talks
09:30 AM	Invited Talk: D. Alvarez-Melis	3:00 PM	Invited Talk: E. L. Dyer
10:00 AM	Coffee Break	3:30 PM	Closing Remarks
10:30 AM	Contributed Talks	3:45 PM	Poster Session
11:30 AM	Mentorship Panel Discussion	5:00 PM	Social Event

workshop proceedings volume. After the success of last year, we organized a *Conference-to-Workshop track* again where we invited a selection of relevant papers from the NeurIPS 2025 main track to be presented in our poster session. 12 papers were accepted on a first-come, first-served basis, provided that they aligned with the workshop’s topics. Additionally, we brought back the *blogpost track*, which received 7 submissions, in which participants were encouraged to present comprehensive guides to known methods and results, opinion pieces, challenges, deep dives, or informal presentations of their own technical submissions in a dynamic format, suitable to host interactive visualization content. These are showcased on the workshop website (following ICLR format).



(a) Submissions and acceptance counts across editions. (b) Accepted papers by submission track.

Figure 2: **UniReps growth and track composition over time.** (Left) Total submissions nearly doubled from II edition to III edition, while maintaining a strong acceptance pipeline. (Right) The accepted set spans multiple contribution formats, reflecting the workshop’s goal of supporting both early-stage and mature work.

As shown in Fig. 2, UniReps has grown steadily over its three editions, from 91 submissions in I Edition to 106 in II Edition (+16%), and to 205 in III Edition (+93% relative to II Edition). Over the same period, we accepted 76, 95, and 140 submissions, respectively (acceptance rates of 84%, 90%, and 68%). Fig. 2 (right) further breaks down accepted papers by track, highlighting the workshop’s multi-format model: extended abstracts and proceedings papers form the core of the program, complemented by the Conference-to-Workshop track and the blogpost track, which broaden participation and encourage diverse forms of contribution beyond traditional papers. All tracks were included in the workshop poster session to allow authors to present their work. A subset of the submissions have been selected for contributed talks during the workshop and the best papers received awards, which are listed below.

Best Paper (Proceedings Track)

Su Hyeong Lee, Richard Ngo
On Defining Neural Averaging.

Best Paper (Extended Abstracts Track)

Yipeng Zhang, Hafez Ghaemi, Jungyoon Lee, Laurent Charlin
Self-Supervised Learning from Structural Invariance.

Honorable Mention (Proceedings Track)

André Longon, David Klindt, Meenakshi Khosla

Superposition Disentanglement of Neural Representations Reveals Hidden Alignment.

Honorable Mention (Extended Abstracts Track)

Daniel Aarao Reis Arturi, Eric Zhang, Andrew Ansah, Kevin Zhu, Ashwinee Panda, Aishwarya Balwani

Shared Parameter Subspaces and Cross-Task Linearity in Emergently Misaligned Behavior.

Program Committee & Chairs

We thank our entire program committee for enabling a smooth reviewing process this year, especially Moshe Eliasof for serving as Area Chair. Our esteemed reviewing committee comprised 281 dedicated reviewers who collectively contributed 521 reviews. Their expertise and commitment were instrumental in ensuring the high quality and rigor of the discussions and findings presented at our workshop. In addition, we presented the Best Reviewer Award to *Artem Kirsanov* and *Zedong Wang* in recognition of their exceptionally thoughtful, rigorous, and constructive reviews. Below is the complete list of reviewers who contributed to the Program Committee for this edition of the workshop.

- Ahmed Qazi (*Tibbling Technologies*)
- Aishwarya Balwani (*St. Jude Research Hospital*)
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- Veddhanth Chakravarthy (*Shiv Nadar University Chennai*)
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- Yiqing Bo (*University of California, San Diego*)
- Yixiao Kang (*Facebook*)
- Zahar Kohut (*Ukrainian Catholic University*)
- Zheyuan Hu (*University of Cambridge*)

Attendance We surpassed our expectations and last year's edition count by drawing in a diverse crowd of over 1000 attendees in person and virtually throughout the day. The audience was a rich tapestry of students, researchers, and industry practitioners from a variety of communities and cultures. The welcoming nature of our event was further enhanced by the thoughtful room setup and environment we created, which fostered a sense of inclusion and engagement among all attendees.

Feedback We ran a feedback survey for the workshop, and below is a summary of the responses. In general, the event received highly positive feedback, with attendees praising the quality of the talks (rated 4.31/5), engaging poster sessions (rated 4/5), and valuable discussions both in person and online. Many appreciated the organization and depth of research presented, with some highlighting the usefulness of platforms like Discord for remote participation (rated 4.8/5). However, there were areas for improvement, including better spacing and navigation for poster sessions, more structured mentorship opportunities, and increased interactivity in sessions given as feedback comments. Some also suggested shortening or revising panels, improving room setups, and incorporating broader topic representation. Despite these points, many participants expressed gratitude to the organizers for a well-executed and insightful event, rating the workshop 4.54/5).

Diversity and Inclusivity Our workshop upheld diversity and inclusivity as fundamental principles for fostering a balanced and productive environment. To achieve this, we strive for diversity in various aspects, including seniority, gender balance, and nationality. Our organizers and invited speakers ranged from PhD students to junior and senior researchers, reflecting a broad spectrum of experience levels. We made a conscious effort to ensure gender balance among both our organizers and keynote speakers, and included participants from different regions, covering Europe, the United States, and

Middle Eastern Asia. To promote an inclusive environment, we actively sought participation from the BlackInAI, Women In Machine Learning (WiML), QueerInAI, and LatinxInAI communities by sending Program Committee calls and invitations to attend the workshop through their mailing lists and communication channels. In this regard, with the generous contribution in funding from the Gatsby and Simons Foundation, we were able to provide financial assistance for attendees from underrepresented groups. This program funded the registration fees for researchers, students, or individuals who encountered financial obstacles when trying to attend NeurIPS and UniReps. In this program, we directly supported 7 students selected by an independent committee, making it more feasible for a broader range of participants to attend and contribute to our workshop.

Sponsors We extend our deepest gratitude to our sponsors, the Gatsby and Simons Foundation, for their generous support and commitment to advancing research and innovation. Their contributions have been invaluable in making our event a success, enabling us to create a platform for sharing knowledge, fostering collaborations, and promoting the latest advancements in the field. We are truly thankful for their support and look forward to continuing our partnership in the future. We additionally thank Mike Sainsbury (Administrative Manager, UCL, Gatsby Unit) his guidance and support in managing the sponsorship process.

Future directions

We find it both crucial and timely to establish a research forum and a supportive community that encourages knowledge exchange at the intersection of machine learning and neuroscience, with a particular emphasis on unified representations. In this spirit, our blog post track remains open for submissions throughout the year, and we actively welcome innovative contributions. As we move forward, we remain dedicated to facilitating meaningful discussions on these topics at NeurIPS and other key events in Machine Learning, Neuroscience and Cognitive Science communities, among others.

Community To strengthen our sense of community, we have also established an active network of students and researchers. This network is a central hub for coordinating activities such as seminars and hackathons, further enriching the UniReps workshop experience. Additionally, we have started a new monthly event, the [ELLIS x UniReps Speaker Series](#) where we investigate UniReps topics. Each session features a keynote and a flash talk, fostering interdisciplinary dialogue across machine learning, neuroscience, and cognitive science. Join us to stay up-to-date with the latest workshop news, connect with a vibrant community, display your latest projects, and remain informed about exciting opportunities, events, and research. Our aim is to foster an engaging and inclusive environment, allowing each participant to contribute, learn, and maintain lasting connections beyond the workshop. Check out the [UniReps Website](#)! In addition, you can follow the last updates on the UniReps community on our [Twitter profile](#), [Bluesky profile](#), and [Discord](#)!

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